
PRODUCT SPECIFICATION

EngCoach

A personalised English-learning system for Bangladeshi learners

Human-authored curriculum. AI-assisted assessment.
Evidence-based learning, explained in Bangla.

Assess → Learn → Practice → Re-assess → Review → Master

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SECTION 01

Problem Statement

English proficiency is a major barrier for many learners in Bangladesh. It affects access to jobs, higher education, scholarships, freelance opportunities, and professional communication.

Many technically capable students and professionals struggle because they cannot confidently:

- Write a professional email or message
- Answer interview questions
- Express ideas clearly in English
- Perform well in written English assessments
- Communicate effectively with clients or colleagues

Existing English-learning solutions often fail to address the specific needs of Bangladeshi learners for three major reasons.

1.1 English Taught Through English

Learners are often given grammar rules, explanations, and error corrections in English—the very language they are struggling to understand. When the explanation itself is unclear, learners memorize rules rather than actually understanding them.

1.2 Lack of Feedback on Output Skills

Reading and grammar can be practiced independently, but writing and speaking require meaningful feedback. Human feedback is often expensive, slow, difficult to access, and unavailable outside major cities. As a result, learners may spend significant time studying English without receiving enough feedback on the skills they actually need for interviews, jobs, academic applications, and professional communication.

1.3 No Evidence of Actual Learning

Many learning platforms measure progress primarily through content consumption. Completing a lesson does not necessarily mean that a learner has understood or mastered the topic. Learners often do not know what they are weak at, whether they have actually improved, which mistakes they repeatedly make, or which topics they should study next.

SECTION 02

Target Users

- University students preparing for job interviews and viva
- Job seekers preparing for written tests and English interviews
- Freelancers who need stronger client communication
- Higher-study and IELTS aspirants
- Working professionals improving workplace English

The platform is particularly focused on learners who benefit from Bangla-medium explanations and Bengali-L1-specific error guidance.

SECTION 03

Solution

EngCoach is a structured English-learning platform where human experts author the curriculum, while AI handles assessment, error analysis, and personalisation.

The platform identifies what a learner already knows, teaches only what they need, evaluates their performance afterward, and continuously tracks their weaknesses and improvement.

The core learning model is:

Assess → Learn → Practice → Re-assess → Review → Master

This transforms English learning from passive content consumption into an evidence-based learning process.

SECTION 04

Core Product Principles

4.1 Human-Authored Curriculum

Learning content, explanations, examples, and educational rules are created and curated by human experts. AI does not independently generate the core curriculum.

4.2 Bangla-Medium Explanation

Rules, mistakes, corrections, and difficult concepts are explained in Bangla where appropriate, using contrastive Bangla-English examples.

4.3 AI-Assisted Assessment

AI is used where it provides significant value at scale, particularly for writing assessment, speaking assessment, error detection, error classification, and personalised learning recommendations.

4.4 Evidence-Based Learning

The platform does not equate lesson completion with mastery. Each topic can show measurable evidence of progress through pre-assessment score, post-assessment score, retention/review performance, and mastery status.

4.5 Personalised Learning

Learners should not spend unnecessary time studying concepts they already understand. The platform identifies weaknesses and adapts the learning path accordingly.

SECTION 05

Navigation Model

The user is not forced through a lengthy onboarding or global placement test.

Primary Navigation

Home → Section → Topic → Learning Flow

Users can choose any section, choose any topic, switch between sections at any time, study topics in any order, and track progress independently for each topic and section.

Each topic establishes the learner's current level through its own pre-assessment.

SECTION 06

Three-Phase Learning Model

PHASE 1

Pre-Assessment

A diagnostic assessment establishes the learner's current ability for a specific topic.

Full Pass

- Learning phase becomes optional
- Topic may be marked as already known

Partial Understanding

- Only weak sub-topics are prioritised

Insufficient Understanding

- Full learning path is recommended

The goal is to identify what the learner does not know, rather than forcing everyone through the same lesson.

PHASE 2

Learning

The learner receives human-authored educational content targeted to the weaknesses identified during pre-assessment.

Content may include:

- Rules
- Examples
- Bangla explanations
- Contrastive Bangla-English examples
- Guided exercises
- Progressive practice

Learning progresses from:

Recognition → Correction → Production

PHASE 3

Post-Assessment

The learner is assessed again to determine whether learning actually occurred.

The platform compares the learner's performance against the pre-assessment.

EXAMPLE

Pre-assessment	42%
Post-assessment	81%
Improvement	+39 percentage points

Pass

- Topic becomes completed
- Improvement is shown
- Topic enters the review/mastery cycle

Fail

- Remaining weak areas are identified
- Targeted remediation is recommended

SECTION 07

Learning Evidence & Mastery

The platform explicitly shows evidence of learning rather than simply marking lessons as completed.

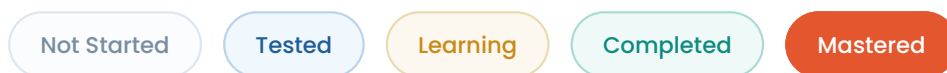
EXAMPLE

Subject-Verb Agreement

Before Learning	42%
After Learning	81%
Retention Review	88%
Status	Mastered

A topic should only receive the Mastered status after demonstrating sustained understanding through later review, rather than simply passing one assessment.

Topic Status



SECTION 08

Personal Learning Intelligence

8.1 Error Notebook

Every meaningful mistake identified during assessment or practice is recorded.

Each error may contain:

- Original error
- Corrected version
- Error type
- Rule
- Bangla explanation
- Relevant learning topic

Errors are periodically resurfaced through spaced review.

The Error Notebook is shared across sections, allowing mistakes from Grammar, Vocabulary, Writing, and Speaking to contribute to the learner's overall profile.

8.2 Personal Weakness Model

The platform aggregates a learner's performance and errors into a personal proficiency profile.

EXAMPLE

Grammar

Articles	54%
Prepositions	61%
Tenses	78%
Subject-Verb Agreement	84%

This model is used to determine:

- What the learner should study
- What they should review
- Which mistakes need reinforcement
- Which topics can be skipped
- Which areas are improving

SECTION 09

Personalised Home

The home screen is not only a navigation page. It acts as the learner's daily learning control center.

It may show:

- Continue Learning
- Weak Areas
- Today's Review
- Daily Goal
- Progress

EXAMPLE

Today's Plan

Review 5 mistakes	5 min
Articles lesson	7 min
Vocabulary review	3 min
Writing practice	5 min

The system should help answer:

"What should I learn today?"

SECTION 10

Feature List

10.1 Core Platform

- OTP-based authentication
- User profile
- Session management
- Logout
- Account deletion
- Subscription status and management
- Section-based home screen
- Continue Learning
- Weak Areas
- Today's Review
- Daily Goal
- Personalised recommendations
- Sections and topic lists
- Topic status
- Topic-level and section-level proficiency
- Independent progress tracking
- Progress dashboard
- Pre-assessment scores
- Post-assessment scores
- Improvement delta
- Retention/review performance
- Time invested
- Weak-area summary

10.2 Grammar

Grammar is organised into individual topics such as Tenses, Articles, Prepositions, Subject-Verb Agreement, Conditionals, Voice, Narration, Modals, Clauses, and Pronouns.

Features

- Topic-specific pre-assessment
- Bangla-medium explanations
- Contrastive Bangla-English examples
- Bengali-L1-specific error patterns
- Progressive drills
- Recognition → Correction → Production practice
- Topic-specific post-assessment
- Targeted remediation
- Error Notebook integration
- Spaced review
- Mastery tracking

10.3 Vocabulary

Vocabulary is organised according to learner level and practical usage.

Features

- Level-banded vocabulary
- Themed vocabulary sets
- Interview, academic, business, and everyday vocabulary
- IELTS-oriented vocabulary
- Bangla meanings
- English definitions
- Example sentences
- Collocations
- Register/context notes
- Confusable-word pairs

- Spaced repetition
- Error Notebook integration

Vocabulary Challenge

- Daily question set
- Streak
- Score
- Leaderboard

10.4 Writing

Writing focuses on practical English output.

Potential task types

- Email
- Paragraph
- Essay
- Report
- Formal letter
- Interview written answer
- Client message
- IELTS-style task

A writing topic may begin with a diagnostic writing task that establishes current proficiency, common error types, structural weaknesses, vocabulary weaknesses, and other relevant weaknesses.

AI Writing Assessment

Writing is evaluated using a fixed, rubric-based assessment system covering:

- Task Response
- Vocabulary
- Coherence
- Mechanics
- Grammar

The rubric uses predefined criteria and anchor examples to improve scoring consistency.

Writing Feedback

- Overall rubric-based score
- Bangla explanation of errors
- Dimension-level scores
- Relevant rule/content reference
- Error classification
- Rewrite suggestion
- Error highlighting
- Stronger-version comparison

Writing History

- Previous attempts
- Improvements
- Scores
- Version comparison
- Errors

Writing errors are added to the Error Notebook.

10.5 Speaking

Version 1 of speaking assessment is transcript-based. The platform does not claim phoneme-level pronunciation assessment in this version.

Prompt types

- Interview question
- Describe a topic
- Opinion
- Situational response
- Read-aloud
- IELTS-style cue card

Flow

Record → Speech-to-Text → Transcript → AI Assessment → Feedback

Assessment

- Fluency
- Speech rate
- Pause frequency and placement
- Filler-word usage
- Grammatical accuracy
- Vocabulary range
- Task completion
- Answer structure

Feedback

- Transcript
- Highlighted errors
- Bangla explanations
- Assessment results
- Model answer
- Comparison with model response

Speaking errors are added to the Error Notebook. A speaking-minute allowance may be applied based on the platform's subscription and operational model.

10.6 Gamification

Gamification is designed to encourage consistent learning rather than replace learning.

Current features

- Daily goals
- Streaks
- Daily vocabulary challenge
- Leaderboard

10.7 Subscription

The platform uses bdapps carrier billing for subscription management.

Free Experience

- Topic pre-assessment
- Diagnostic result
- Limited sample learning content

Paid Experience

- Complete learning content
- AI assessments
- Full progress tracking
- Error Notebook
- Personalised learning
- Speaking practice
- Writing assessment
- Vocabulary practice

Usage limits for computationally expensive AI features will be determined after real-world product usage and cost measurement.

10.8 Privacy & User Data

The platform handles potentially sensitive learning data, including:

- Writing submissions
- Speaking transcripts
- Learning history
- Assessment results
- Error history
- Proficiency information

The product should follow a privacy-conscious data policy and clearly communicate:

- What user data is collected
- Why it is collected
- How it is processed
- How long it is retained
- How users can delete their data

Raw speech recordings should not be retained indefinitely unless there is a clear product requirement for doing so.

SECTION 11

Future Features

A. Reading Section

A dedicated Reading section with level-based and topic-based reading materials, comprehension practice, vocabulary support, and reading assessment. It will follow the same Pre-Assessment → Learning → Practice → Post-Assessment → Mastery model.

B. Listening Section

A dedicated Listening section with level-based audio content, transcripts, comprehension practice, vocabulary support, and listening assessment. It will follow the same Pre-Assessment → Learning → Practice → Post-Assessment → Mastery model.

C. Advanced Learning

- Cross-skill personalised learning
- Adaptive daily learning plan
- Intelligent spaced repetition
- Cross-skill Error Notebook

D. Advanced Speaking

- Phoneme-level pronunciation assessment
- Bengali-L1 pronunciation analysis
- Advanced pronunciation and fluency feedback

E. Gamification & Social

- Challenge a Friend
- Real-Time 1v1 Vocabulary Matches
- Section-Specific Leaderboards

F. Progress & Certification

- Shareable Proficiency Report
- Skill-Specific Reports
- Overall English Proficiency Score
- English Proficiency Certification
- Shareable Learning Profile

SECTION 12

What This Product Is Not

Not a Chatbot Wrapper

The platform is not primarily a chatbot with English-learning prompts. The curriculum is human-authored and structured.

Not a Generic Global English App

The platform is designed specifically around the needs of Bangladeshi learners, including Bangla explanations and Bengali-L1-specific error patterns.

Not Passive Content Consumption

Completing lessons does not automatically mean mastery. The platform measures learning through assessment, improvement, retention, and mastery.

Not AI-Generated Curriculum

AI supports assessment and personalisation, while the core educational content remains human-authored and curated.

SECTION 13

Core Product Loop

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|----|------------------|--|
| 01 | Assess | Identify what the learner knows and does not know. |
| 02 | Learn | Provide targeted, human-authored instruction with Bangla explanations. |
| 03 | Practice | Allow the learner to apply the concept. |
| 04 | Re-assess | Measure whether learning occurred. |
| 05 | Review | Resurface mistakes and weak areas through spaced repetition. |
| 06 | Master | Confirm sustained understanding through later performance. |
| 07 | Prove | Show measurable evidence of improvement. |

SECTION 14

Product Vision

EngCoach aims to become a personalised English-learning system for Bangladeshi learners where every learner can understand:

What I don't know → Why I am making mistakes → What I should learn →
Whether I improved → What I should review next.

The goal is not simply to help users consume more English content. The goal is to help them build measurable English proficiency and see evidence that they are improving.